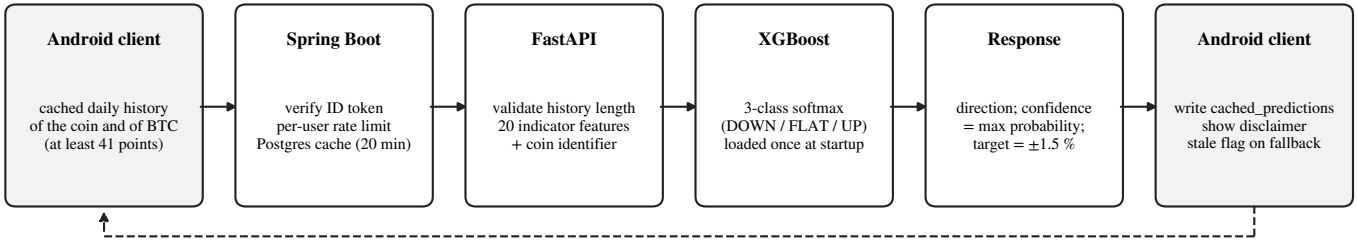
Android client
cached daily history
of the coin and of BTC
(at least 41 points)
Spring Boot
verify ID token
per-user rate limit
Postgres cache (20 min)
FastAPI
validate history length
20 indicator features
+ coin identifier
XGBoost
3-class softmax
(DOWN / FLAT / UP)
loaded once at startup
Response
direction; confidence
= max probability;
target = ±1.5 %
Android client
write cached_predictions
show disclaimer
stale flag on fallback
on a failed call or a rate-limit response, the client shows the cached prediction instead